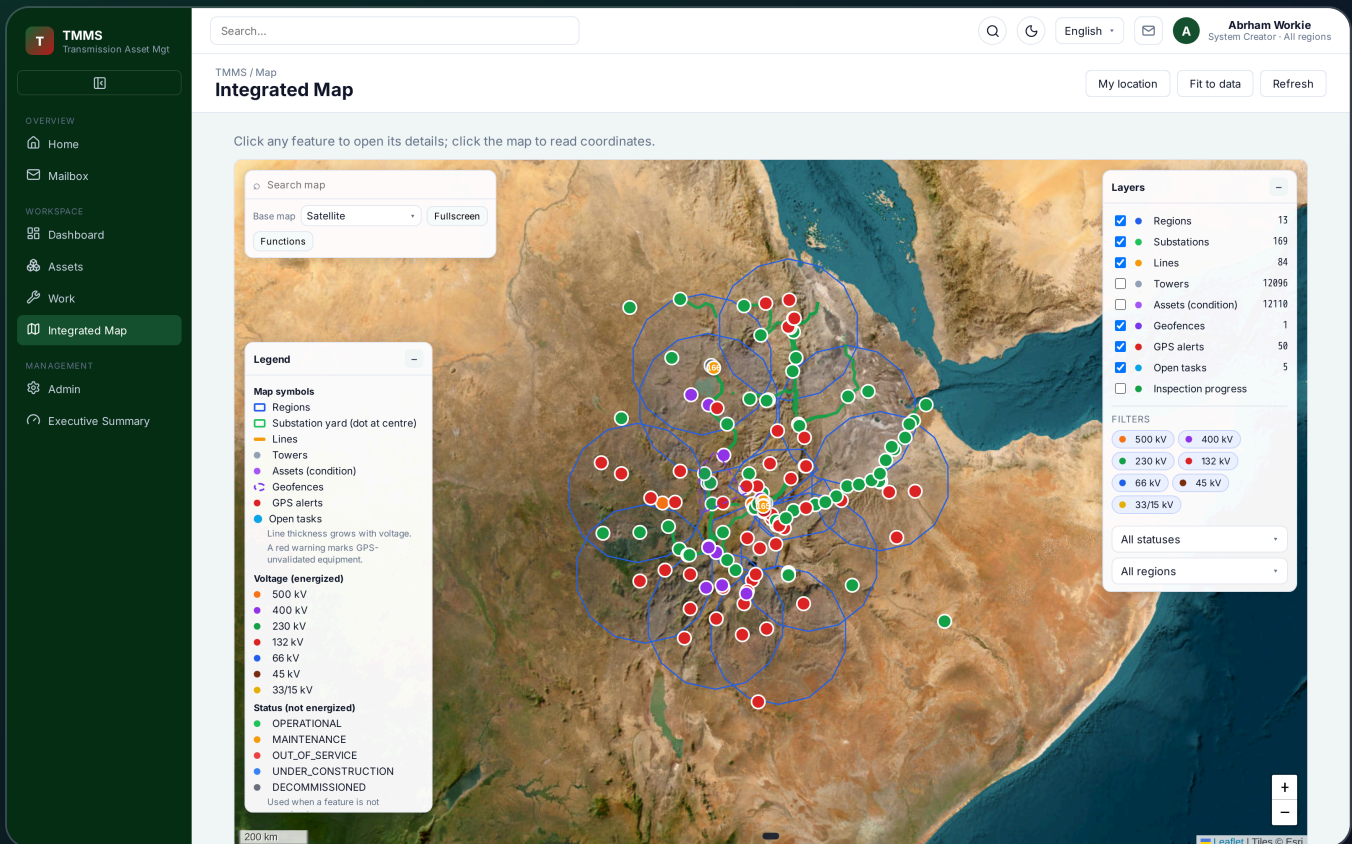


Every tower. Every task. Every proof.

TMMS is a geographic, asset-centric operations platform for high-voltage transmission networks. It keeps one authoritative record of every asset and location, and drives each piece of work through a closed loop: **Plan** → **Dispatch** → **Execute** → **Document** → **Verify**.

**12,148**

Georeferenced assets

13

Regions managed

6

Lifecycle stages

1

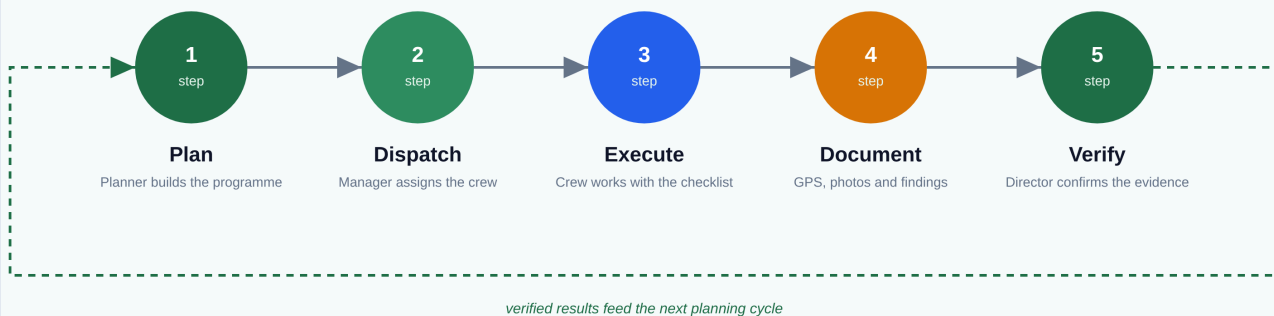
Source of truth

One closed loop from plan to proof

Utilities rarely fail for lack of work orders. They fail because the plan, the dispatch, the field evidence and the verification live in separate places — so nobody can prove what happened, where, or when. TMMS closes that loop in a single georeferenced system.

The closed maintenance loop

Every piece of work moves through five accountable stages — and feeds the next plan.



1

Georeferenced by default

Every substation, line, tower and asset carries coordinates. Locations are verified, not trusted.

2

Accountable to the person

Each stage has one owner. The verifier can never be the executor, and every action is logged.

3

Nothing disappears

Recurring work is automated, evidence is retained, and closed work feeds the next plan.

Built for the field, trusted by the board

A crew sees only its assigned work. A manager sees the crews it commands. A director sees the region. An executive sees the whole network — the same data, filtered by role, never duplicated.

THE COMMAND CENTER

A home that tells each role exactly what to do

Every signed-in user gets a function banner and one card per task bucket their role owns — **execute, schedule, assign, verify** or **review**. Buckets are separate sections, never merged into one queue, so scheduling, assignment, execution and verification never mix.

TMMS
Transmission Asset Mgt

OVERVIEW

WORKSPACE

MANAGEMENT

Search...

English

Abraham Workie
System Creator - All regions

TMMS / Home

Home

Welcome, Abraham Workie

All regions

System administration

Corporate HQ

System map

YOUR WORK BY CATEGORY

Pick a category tab

To schedule 2

To assign 1

Assigned to your crews 73

To verify 0

Emergency 1

Review evidence 0

Oversight 0

Assigned to your crews

Work already handed to crews you command — an oversight lens on active assignments.

Search...

Group by

Status

Type

Substation

Line

Asset category

In progress

5

TK-2026-000050

In progress

bbb — C1-001-004

Inspection - Medium - C1-001-004

Overdue

08/11/2026

Submit for verification

TK-2026-000051

In progress

bbb — C1-001-005

Inspection - Medium - C1-001-005

Overdue

08/11/2026

Submit for verification

TK-2026-000067

In progress

bbb — C1-001-021

Inspection - Medium - C1-001-021

Overdue

08/11/2026

Submit for verification

TK-2026-000147

In progress

Tower Inspection — C1-0012-032

Preventive - Medium - C1-0012-032

Overdue

09/30/2026

Submit for verification

TK-2026-000111

In progress

Multi-checklist smoke test

Preventive - Medium —

Overdue

10/05/2026

Submit for verification

Assigned

67

On hold

1

HISTORY

Mine

In my area

Home — role-scoped buckets with live counts and collapsible sub-categories.

TMMS
Transmission Asset Mgt

Dashboard

Search...

English

Abraham Workie
System Creator - All regions

All regions

Generated: 10/05/2025, 01:55 PM

Export view

ASSETS IN SERVICE

12,148

12,148 assets

MEAN CONDITION

8

100% of 12,148 assessed

OPEN WORK ORDERS

77

10 overdue

INVENTORY SPEND

\$378,296

100% of 12,148 assets

CREW READINESS

92%

20% of 12,148 assets

Condition distribution

12,148 assets

12,148 assets

Degradation attention

Top 5 by negative movement

View all

DEMO-C1-001-01

10/05/2025

Central 1 Region Transmission Substation Operation - suggested 270

-7

DEMO-C1-001-01

10/05/2025

Central 1 Region Transmission Substation Operation - suggested 270

-6

ART-TX-001

10/05/2025

Central 1 Region Transmission Substation Operation - suggested 470

-3

ART-C2-001

10/05/2025

Central 2 Region Transmission Substation Operation - suggested 170

-3

TWR-C2-001-021

10/05/2025

Central 2 Region Transmission Substation Operation - suggested 070

-2

12-month maintenance spend

Total: \$378,296

Latest: 7/3/20

Condition trend

Latest: 7/3/20

Dashboard — operational overview.

TMMS
Transmission Asset Mgt

Organization

Search...

English

Abraham Workie
System Creator - All regions

How the organization maps to the system

Each bucket has a function banner and a card per task bucket their role owns — execute, schedule, assign, verify or review. Buckets are separate sections, never merged into one queue, so scheduling, assignment, execution and verification never mix.

Corporate / Business Unit - Executive — oversight
Regional Director - Single Director — oversight
Department - Department Manager — assign & verify
Substation Unit - Substation Manager — assign & verify
Asset Unit - Asset Unit Manager — oversight
Planning and Dispatch Unit - Planner / Dispatcher — schedules / assigns

ALL UNITS

27

13

16

12

HIERARCHY (CEO — TEU — DIVISIONS — DIRECTORATES — DEPARTMENTS)

ALL UNITS

27 of 27

CODE

NAME

TYPE

FUNCTION

REGION

MANAGER

CREW

HE-001

Chief Executive Officer

Corporate HQ

Executive oversight

—

Abraham Workie

0

HE-101

Transmission Business Unit

Business Unit

Executive oversight

—

Davidson Harte

0

HE-102

Substation Operation

Division

Executive oversight

—

Olivia Zeleke

0

HE-103

Regional Director

Division

Executive oversight

—

Wenderson Akene

1

HE-104

Central 1 Region Transmission Substation Operation

Regional Directorate

Oversee the region - assign & verify

C1

Dennis Gwangwa

1

HE-105

Central 2

Division

Oversee the region - assign & verify

—

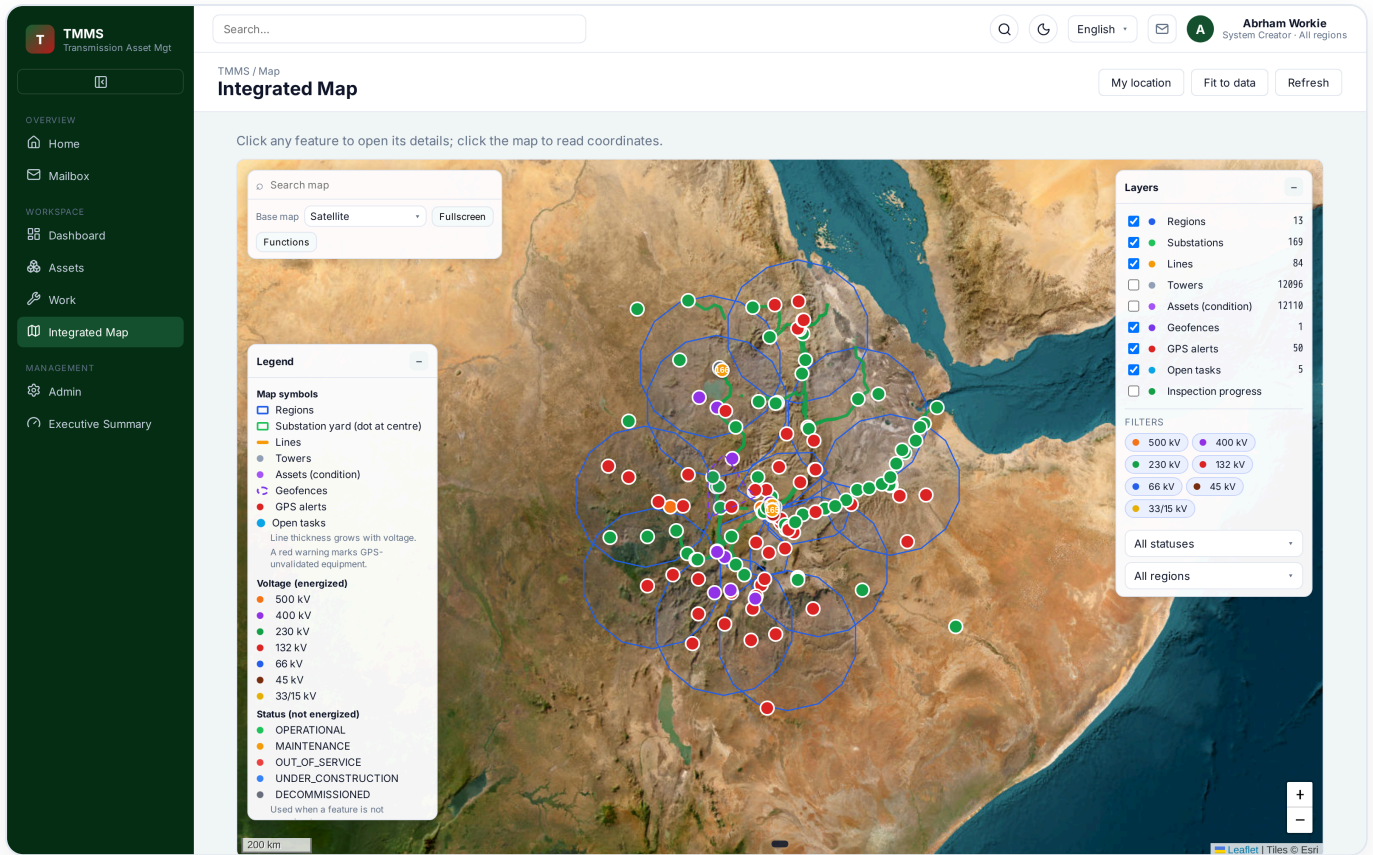
Dennis Gwangwa

1

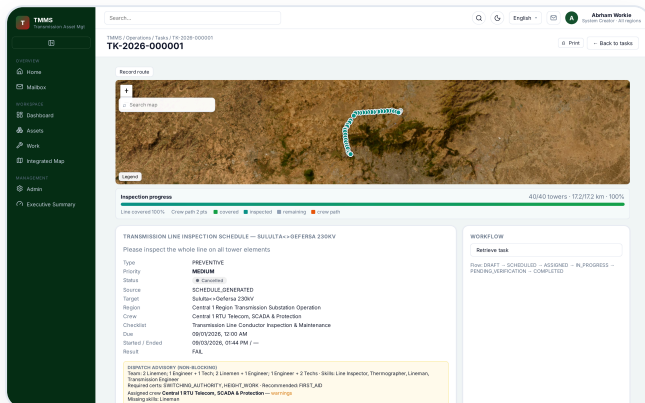
Organization — each unit's function and the role that heads it.

See the whole network, then prove you were there

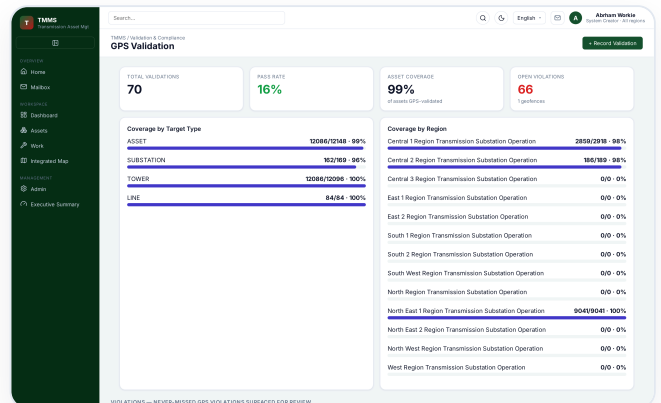
Satellite and vector layers plot substations, transmission lines and towers with live inspection coverage. Field capture records a GPS point against the expected device location, so a completed step carries proof of presence — not just a claim.



Integrated map — substations, line corridors and asset coverage in one view.



Task route — line coverage, crew path and progress.



GPS validation — capture against device tolerance.

Geofences

Coordinates are checked against the real world before they are trusted.

Coverage

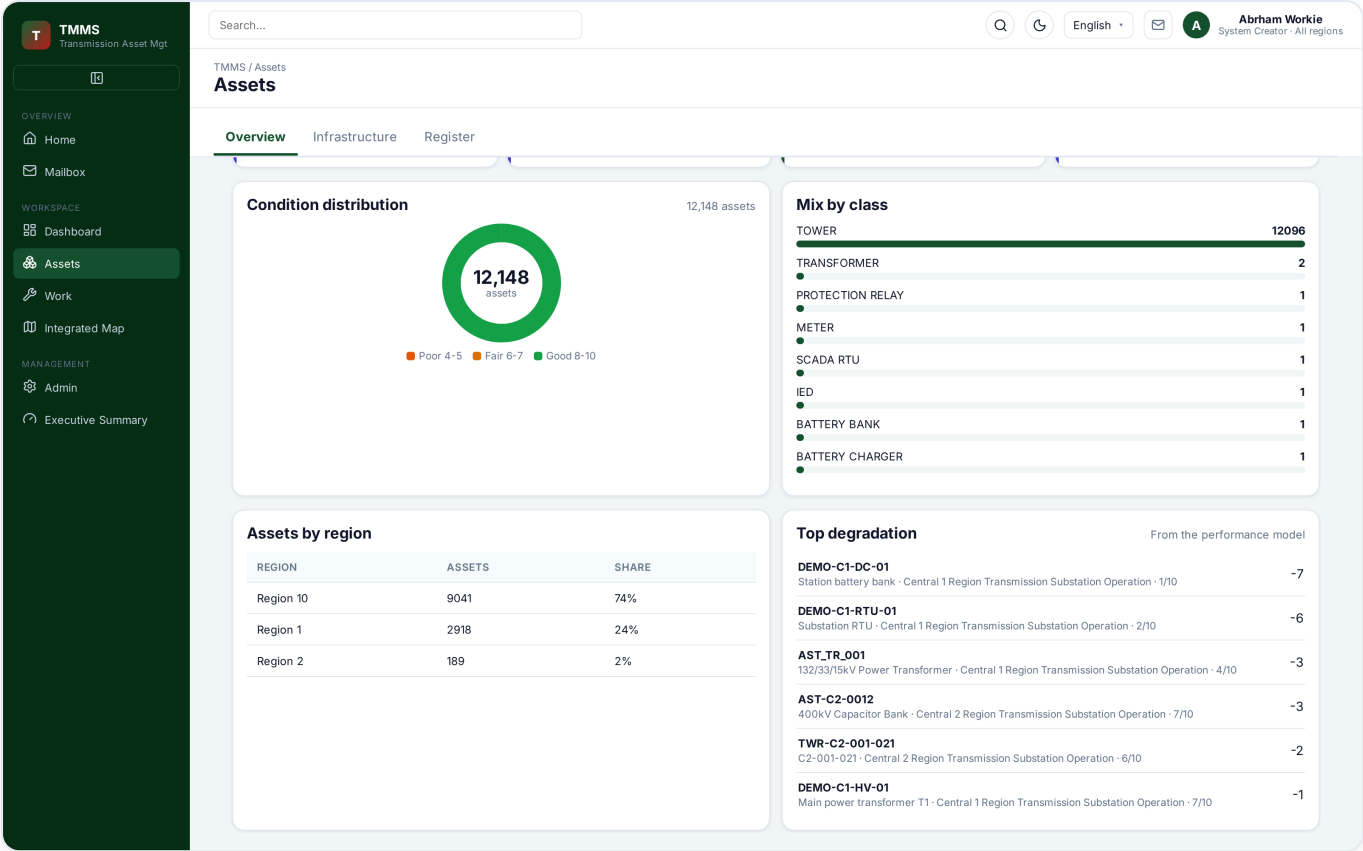
Line and tower inspection progress computed automatically.

Violations

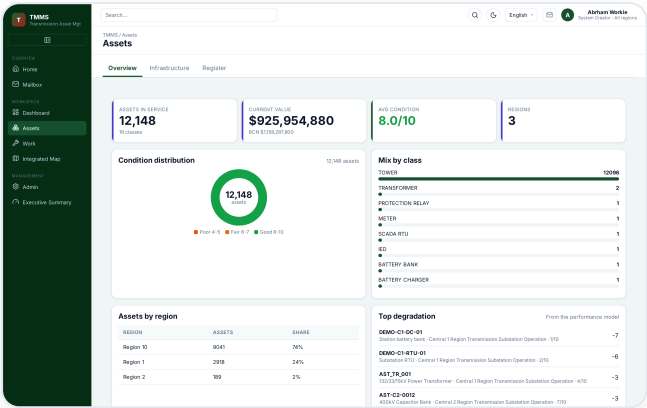
Out-of-tolerance captures are reviewable and auditable.

One authoritative record of every asset

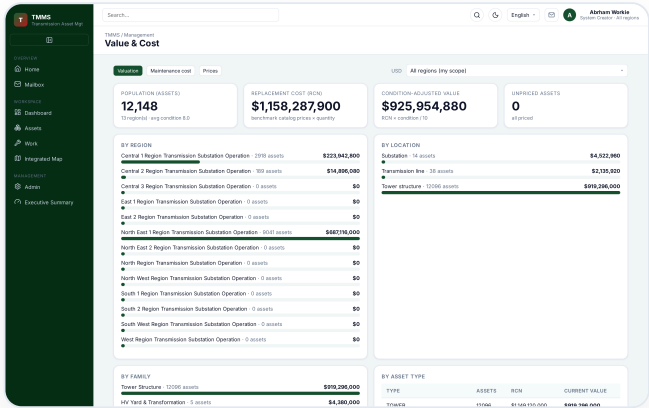
Regions, substations, transmission lines, towers, components and equipment are modelled as one connected tree. Condition scores, class mix, per-region distribution and the top degrading assets are computed from the same source of truth that drives the map and the work.



Assets overview — condition distribution, mix by class, by region and top degradation.



Asset register — paged, filtered, exportable.



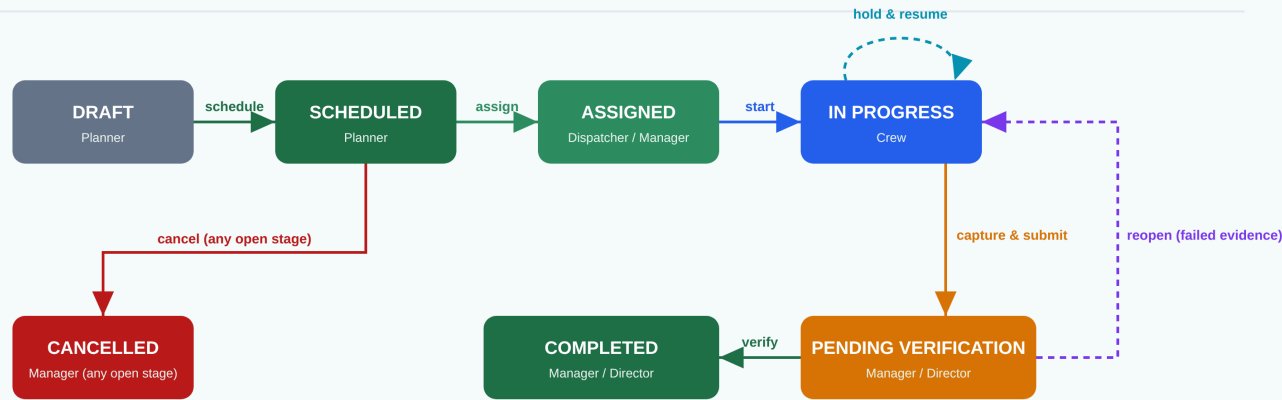
The same register carries replacement cost and condition-adjusted value.

A lifecycle with one owner at every step

Work moves through six accountable stages and cannot skip one. Checklist or GPS gates block completion until the evidence is complete, and the verifier can never be the person who executed the task.

Task lifecycle: one owner at every step

A task cannot skip a stage, and the verifier can never be the person who executed the task.



TASK	TYPE	PRIORITY	STATUS	PROGRESS	TARGET	DUE	CREW	ACTION
TK-2026-000001	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000002	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000003	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000004	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000005	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000006	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000007	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		
TK-2026-000008	Transmission Line Inspection	Schedule	Substation	230kV	08/03/2026	Central 1 RTU Telecom, SCADA & Protection		

Task list — full list with filtering and bulk work.

EXECUTION	CREW	EXECUTED BY	TEMPLATE (S)	SUBMITTED	RESULT	STOPS (PASS/FAIL)	NOTES
E1-0001	Central 1 RTU Telecom, SCADA & Protection	Adhish Bahra	Transmission Line Conductor Inspection & Maintenance (S)	08/04/2026, 07:32 AM	OK	65	
E1-0002	Central 1 RTU Telecom, SCADA & Protection	Adhish Bahra	Transmission Line Conductor Inspection & Maintenance (S)	08/04/2026, 07:30 AM	OK	62	

Task evidence — checklist executions, photos, signature and document.

Emergency path

A task can be raised straight to ASSIGNED, skipping draft and calendar.

Follow-up work

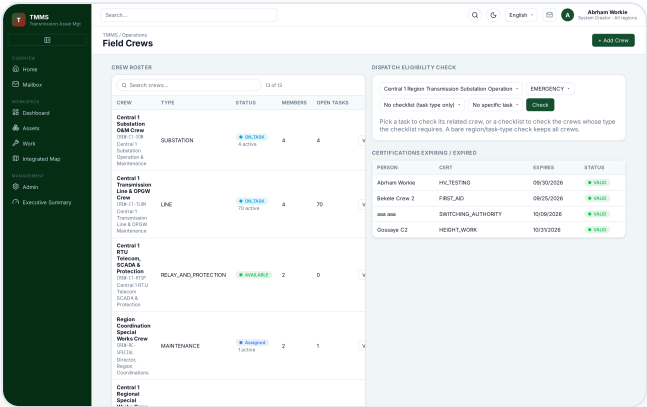
Failed evidence auto-suggests corrective or emergency follow-ups.

Dispatch eligibility

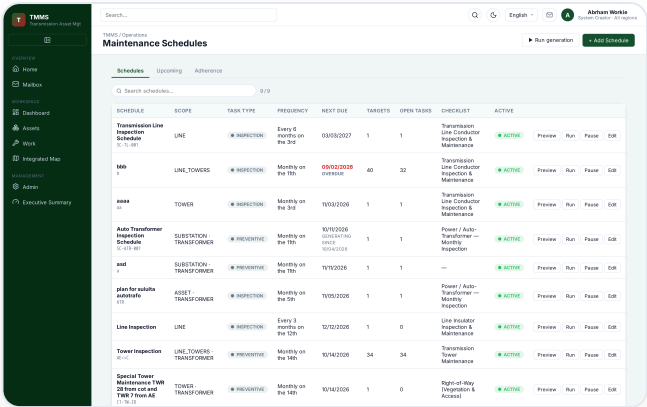
Skills and certifications are checked before a crew is assigned.

Crews, schedules, checklists and certifications

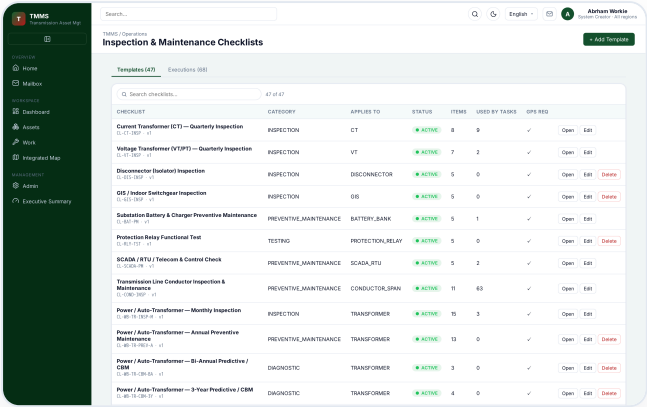
Recurring maintenance is planned once and generated automatically. Crews execute standardised checklists step by step, and the platform knows who holds which qualification and when it expires.



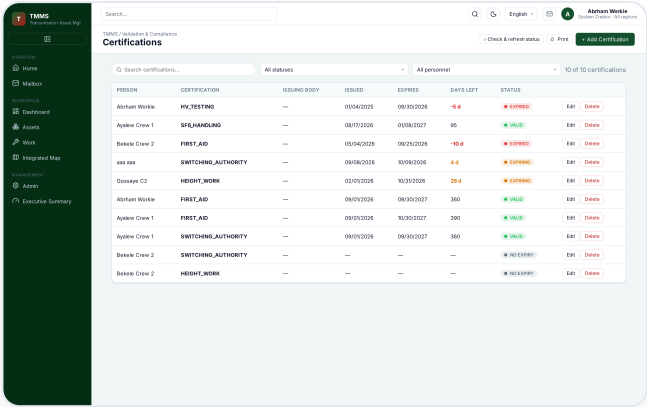
Crews — members, skills and certifications.



Schedules — recurrence rules and generation.



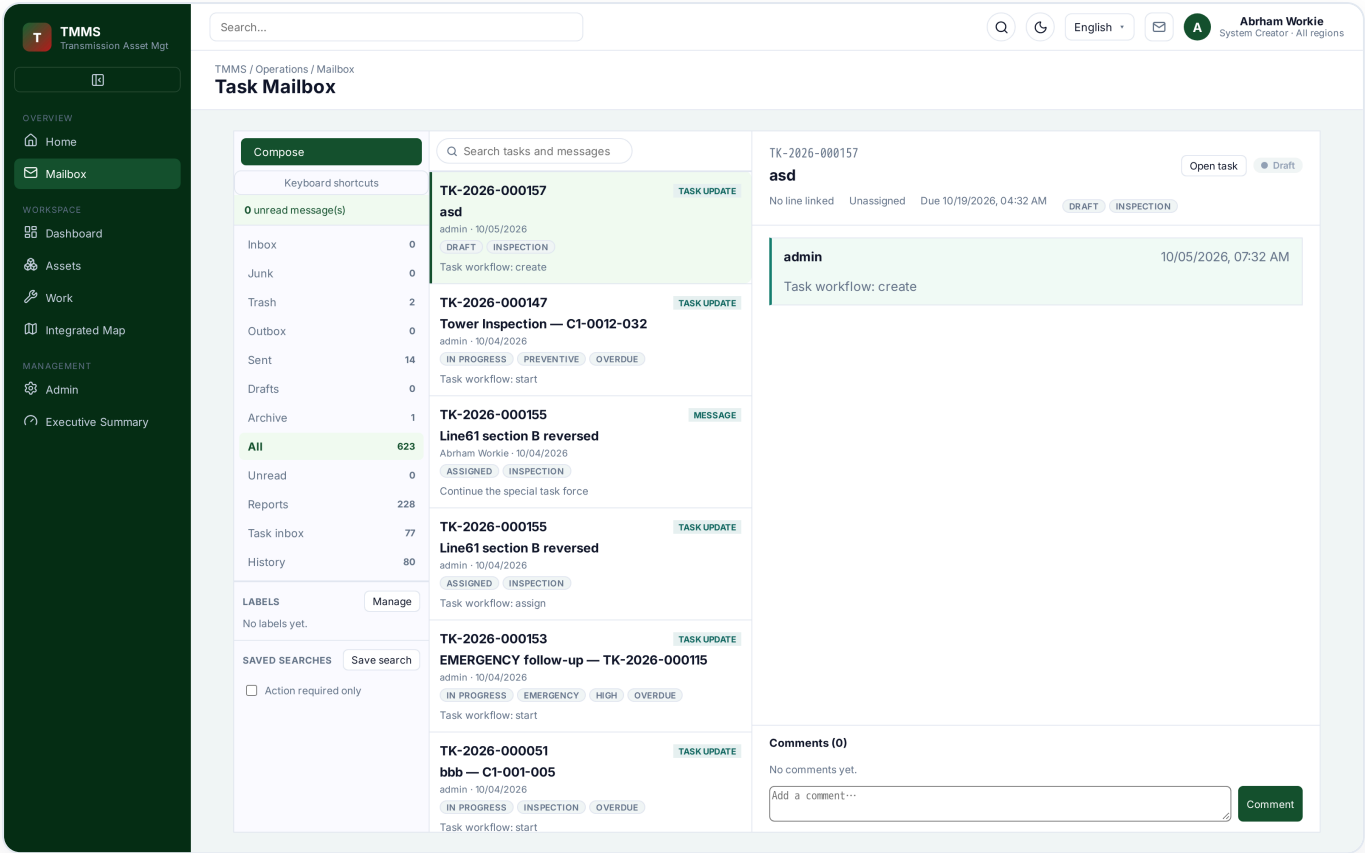
Checklists — templates and step-by-step execution.



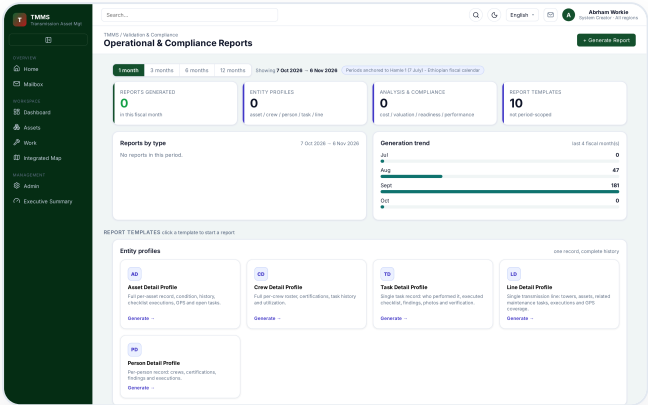
Certifications — qualification and expiry tracking.

Every task update, message and report in one inbox

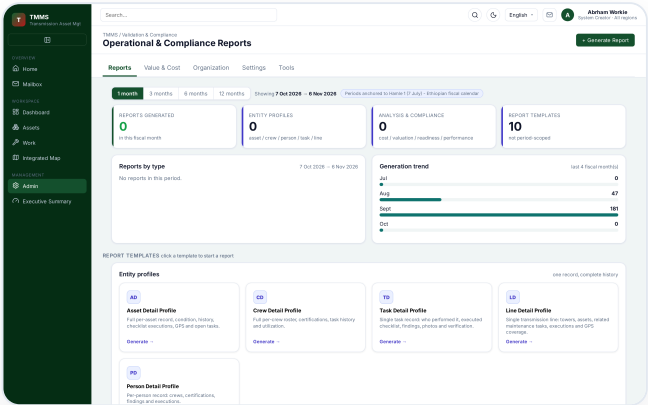
The mailbox is the collaboration spine of the operation: task updates, direct messages and generated reports arrive as a single, searchable activity stream, with folders, labels and saved searches.



Task mailbox — 623 activity records across folders, labels and saved searches.



Reports — operational and compliance outputs.

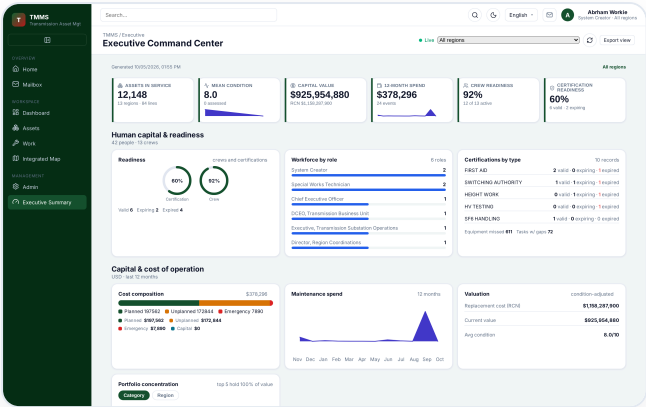


Administration — reports, users and governance.

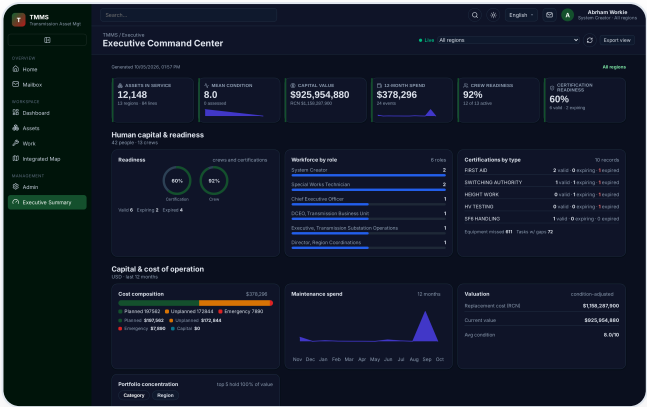
OVERSIGHT

Executive clarity, organisational discipline

An executive summary distils critical overdue work, GPS violations, expiring certifications and cost into a single board-ready view. The organisation model keeps each unit's function and the role that heads it explicit — so authority is never ambiguous.



Executive summary — risks, compliance and value.



Dark theme for control-room and after-hours review.

One function per role

The organisation, the account role and the task bucket are three faces of the same model.

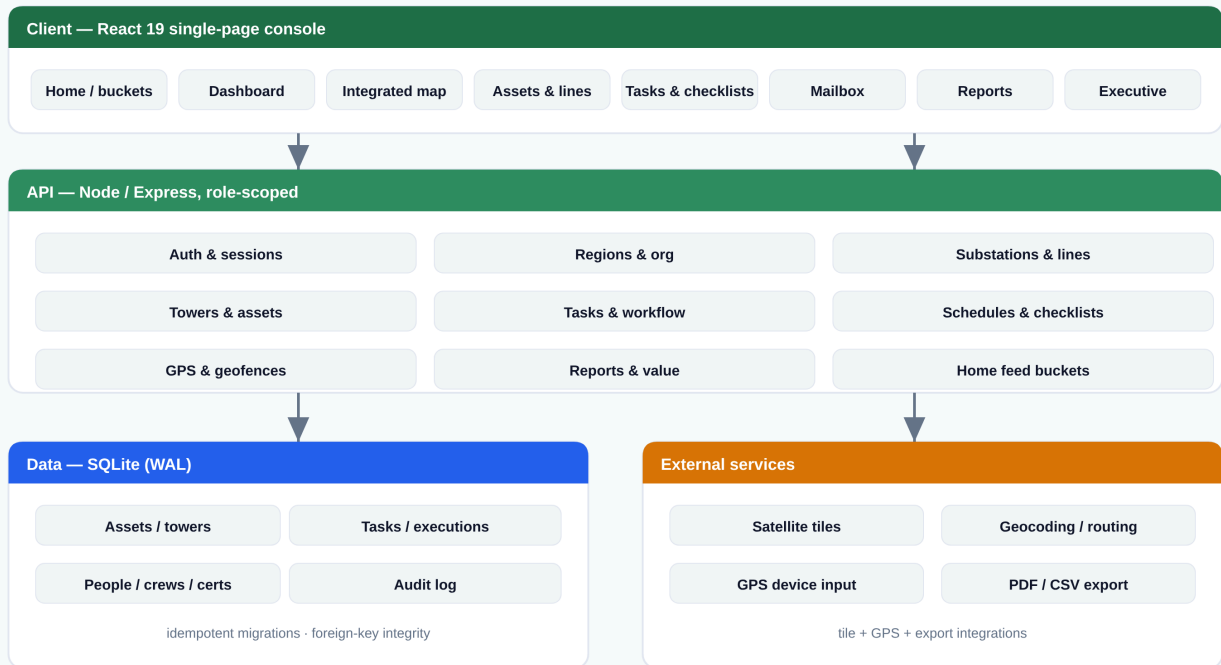
Role	Function	Leads with
EXECUTIVE	Company-wide oversight	Oversight
REGION_DIRECTOR	Owens the region	Assign + Verify
REGION_MANAGER	Commands the department	Assign + Verify
SUBSTATION_MANAGER	Commands the substation	Assign + Verify
PLANNER	Plans the programme	Schedule
DISPATCHER	Dispatches the work	Assign
CREW_LEAD / CREW_MEMBER	Executes the work	Execute
AUDITOR	Independent review	Review evidence

Modern, secure and built to stay consistent

A React single-page console talks to a role-scoped Node / Express API over one authoritative SQLite datastore. Schema changes are idempotent, referential integrity is enforced, and every mutation is written to an audit log — with a redacted, size-capped request body.

System architecture

A single-page operations console over a geo-aware API and one authoritative datastore.



Role-scoped reads

Queries are filtered by region and department, so a manager only ever sees its own domain.

Referential integrity

Cascades are transactional and leave `foreign_key_check` empty.

Auditable by design

Sessions expire, expired sessions are pruned, and sensitive fields are redacted.

TRANSMISSION ASSET MANAGEMENT SYSTEM

One platform. One truth. Every tower, task and proof.

Plan · Dispatch · Execute · Document · Verify